QQ-A-200/17A
14 July 1983
SUPERSEDING
QQ-A-200/17
March 11, 1973

FEDERAL SPECIFICATION SHEET

ALUMINUM ALLOY 6162, BAR, ROD, SHAPES, TUBE AND WIRE, EXTRUDED

This specification was approved by the Assistant Administrator, Office of Federal Supply and Services, General Services Administration, for the use of all Federal agencies.

The complete requirements for procuring the aluminum alloy 6162 bar, rod, shapes, tube and wire, extruded described herein shall consist of this document and the latest issue of QQ-A-200/GEN (see 2.1).

1. SCOPE AND CLASSIFICATION

1.1 Scope. This specification covers the specific requirements for aluminum alloy 6162 bar, rod, shapes, tube and wire produced by extrusion.

1.2 Classification.

1.2.1 Tempers. The bar, rod, shapes, tube and wire shall be of the following tempers, as specified (see 6.2): T5, T5510, T5511, T6, T6510 and T6511. The definition of these tempers shall be as specified in QQ-A-200/GEN.

1.2.2 Tubing. Tubing shall be additionally classified as follows:

Type	Description
I	Tubing extruded from hollow billets using die and mandrel (see QQ-A-200/GEN)
II	Tubing extruded from solid billets using a porthole or spider die or similar tooling (see QQ-A-200/GEN)

2. APPLICABLE DOCUMENTS

2.1 Government publications. The issues of the following documents, in effect on date of invitatin for bids or solicitation for offers, form a part of this specification to the extent specified herein.

FSC 9530, 9525,
4710, 9540

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T12:04Z
Check the source to verify that this is the current version before use.

Federal Specifications

QQ-A-200/GEN - Aluminum Alloy, Bar, Rod, Shapes, Structural Shapes, Tube and Wire, Extruded; General Specification for

(Activities outside the Federal Government may obtain copies of Federal specifications, standards, and commercial item descriptions, as outlined under General Information in the Index of Federal Specifications, Standards and Commercial Item Descriptions. The Index, which includes cumulative bimonthly supplements as issued, is for sale on a subscription basis by the Superintendent of Documents, U. S. Government Printing Office, Washington, DC 20402.

(Single copies of this specification and other Federal specifications and commercial item descriptions required by activities outside the Federal Government for bidding purposes are available without charge from General Services Administration Business Service Centers in Boston, MA; New York, NY; Philadelphia, PA; Washington, DC; Atlanta, GA; Chicago, IL; Kansas City, MO; Fort Worth, TX; Houston, TX; Denver, CO; San Francisco, CA; Los Angeles, CA; and Seattle, WA.

(Federal Government activities may obtain copies of Federal standardization documents and the Index of Federal Specifications, Standards, and Commercial Item Descriptions from established distribution points in their agencies.)

Military Standard

MIL-STD-2154 - Inspection, Ultrasonic, Wrought Metals, Process for

(Copies of military specifications and standards required by contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

3. REQUIREMENTS

3.1 Chemical composition. The chemical composition shall conform to the requirements specified in table I.

3.2 Mechanical properties.

3.2.1 Mechanical properties of material as supplied. The mechanical properties in the direction of extrusion shall conform to the requirements specified in table II.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T12:04Z
Check the source to verify that this is the current version before use.

TABLE I. Chemical composition 1/

Element	Percent	
	Minimum	Maximum
Zinc	--	0.25
Magnesium	0.7	1.1
Copper	--	0.20
Chromium	--	0.10
Manganese	--	0.10
Iron	--	0.50
Silicon	0.40	0.8
Titanium	--	0.10
Other elements, each	--	0.05
Other elements, total 2/	--	0.15
Aluminum	Remainder	

1/ Analysis shall routinely be made only for the elements specifically mentioned in table I. If, however, the presence of other elements is indicated or suspected in amounts greater than the specified limits, further analysis shall be made to determine that these elements are not present in excess of specified limits

2/ The sum of those "Others" metallic elements 0.010 percent or more each, expressed to the second decimal before determining the sum.

TABLE II. Mechanical properties

Temper	Thickness, (bar and shapes); diameter, (rod and wire); wall thickness, (tube); inches	Tensile strength,	Yield strength at 0.2 percent offset or at extension indicated		Elongation in 2 inches or 4 times D 1/, 3/
		Minimum, ksi	Minimum, ksi	Extension under load, inch/inch	Minimum, percent
T5, 2/ T5510, T5511	Up thru 1.000	37.0	34.0	0.0054	7
T6, 2/ T6510, T6511	Up thru 0.249 0.250 - 0.499	38.0 38.0	35.0 35.0	0.0053 0.0053	8 10

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T12:04Z
Check the source to verify that this is the current version before use.

TABLE II. Mechanical properties - (CON)

1/ D represents specimen diameter
2/ For stress-relieved tempers, the characteristics and properties, other than those specified, may differ somewhat from the corresponding characteristics and properties of material in the basic temper
3/ See QQ-A-200/GEN for elongation requirement exceptions

3.3 Internal soundness. When specified (see 6.2), extrusions shall be subjected to ultrasonic inspection in accordance with 4.2. Standards for acceptance shall be as agreed upon by the contractor and the procuring activity.

3.4 Marking. In addition to the marking required by QQ-A-200/GEN, material in the T6, T6510 and T6511 tempers shall be identified by a lot number marked in at least one location on each piece.

4. QUALITY ASSURANCE PROVISIONS (see QQ-A-200/GEN)

4.1 Lot. An inspection lot shall consist of an identifiable quantity of bar, rod, shapes, tube or wire of the same alloy, temper, section thickness and size subjected to inspection at one time.

4.2 Ultrasonic inspection method. The method of ultrasonic inspection shall conform to MIL-STD-2154 and be as agreed upon by the procuring activity and the contractor (see 6.2).

5. PREPARATION FOR DELIVERY (see QQ-A-200/GEN)

6. NOTES

6.1 Intended use. This alloy is intended for use where a weldable, moderate strength alloy having good corrosion resistance is required.

6.2 Ordering data. Purchasers should select the preferred options permitted herein, and include the following information in procurement documents:

(a) Title, number and date of this specification
(b) Form, quantity and temper required (see 1.2.1)
(c) Dimensions required
(d) Requirements for sizes not specifically covered (see QQ-A-200/GEN)
(e) Special end use requirements
(f) Whether type 1 tubing is required (see 1.2.2)
(g) For military agencies - selection of applicable levels of preservation, packaging and packing required. Also specify marking requirements (see section 5, QQ-A-200/GEN)

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T12:04Z
Check the source to verify that this is the current version before use.

(h) For civil agencies - specify packing level, level A or commercial. Also specify marking requirements (see section 5, QQ-A-200/GEN)
(i) Whether ultrasonic inspection of extrusions is required (see 3.3), and, if so, the acceptance limits and method of inspection

MILITARY INTEREST:

Custodians

Army-MR
Navy-AS
Air Force-20

Review Activities

Air Force-99
DLA-IS

CIVIL AGENCY COORDINATING ACTIVITIES:

GSA-FSS
NBS

PREPARING ACTIVITY:

NAVY-AS

DOD Project 9530-0235

Orders for this publication are to be placed with the General Services Administration, acting as an agent for the Superintendent of Documents. See Section 2 of this specification to obtain extra copies and other documents referenced herein.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T12:04Z
Check the source to verify that this is the current version before use.

INSTRUCTIONS: In a continuing effort to make our standardization documents better, the DoD provides this form for use in submitting comments and suggestions for improvements. All users of military standardization documents are invited to provide suggestions. This form may be detached, folded along the lines indicated, taped along the loose edge *(DO NOT STAPLE)*, and mailed. In block 5, be as specific as possible about particular problem areas such as wording which required interpretation, was too rigid, restrictive, loose, ambiguous, or was incompatible, and give proposed wording changes which would alleviate the problems. Enter in block 6 any remarks not related to a specific paragraph of the document. If block 7 is filled out, an acknowledgement will be mailed to you within 30 days to let you know that your comments were received and are being considered.

NOTE: This form may not be used to request copies of documents, nor to request waivers, deviations, or clarification of specification requirements on current contracts. Comments submitted on this form do not constitute or imply authorization to waive any portion of the referenced document(s) or to amend contractual requirements.

(Fold along this line)

(Fold along this line)

DEPARTMENT OF THE NAVY
Commanding Officer
Naval Air Engineering Center
Engineering Specifications and Standards Department
(ESSD), Code 93
Lakehurst, NJ 08733

OFFICIAL BUSINESS
PENALTY FOR PRIVATE USE $300

NO POSTAGE NECESSARY IF MAILED IN THE UNITED STATES

BUSINESS REPLY MAIL
FIRST CLASS PERMIT NO. 12503 WASHINGTON D C

POSTAGE WILL BE PAID BY THE DEPARTMENT OF THE NAVY

Commanding Officer
Naval Air Engineering Center
Engineering Specifications and Standards Dept.
(ESSD), Code 93
Lakehurst, NJ 08733

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T12:04Z
Check the source to verify that this is the current version before use.

STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

(See Instructions - Reverse Side)

1. DOCUMENT NUMBER	2. DOCUMENT TITLE
QQ-A-200/17A	Aluminum Alloy 6162, Bar, Rod, Shapes, Tube and Wire, Extruded

3a. NAME OF SUBMITTING ORGANIZATION	4. TYPE OF ORGANIZATION *(Mark one)*
	☐ VENDOR ☐ USER ☐ MANUFACTURER ☐ OTHER *(Specify)*: _____
b. ADDRESS *(Street, City, State, ZIP Code)*	

5. PROBLEM AREAS

a. Paragraph Number and Wording:

b. Recommended Wording:

c. Reason/Rationale for Recommendation:

6. REMARKS

7a. NAME OF SUBMITTER *(Last, First, MI)* – Optional	b. WORK TELEPHONE NUMBER *(Include Area Code)* – Optional
c. MAILING ADDRESS *(Street, City, State, ZIP Code)* – Optional	8. DATE OF SUBMISSION *(YYMMDD)*

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T12:04Z
Check the source to verify that this is the current version before use.